

$$\int_0^\infty \frac{\sin^2 x \cos x}{e^x} dx$$

$$\frac{1}{2} \int_0^\infty \frac{(1 - \cos 2x) \cos x}{e^x} dx$$

$$\frac{1}{2} \int_0^\infty \frac{\cos x - \cos x \cos 2x}{e^x} dx$$

$$\frac{1}{2} \int_0^\infty \frac{\cos x - \frac{1}{2}(\cos 3x + \cos x)}{e^x} dx$$

$$\frac{1}{2} \int_0^\infty \frac{\frac{1}{2} \cos x - \frac{1}{2} \cos 3x}{e^x} dx$$

$$\frac{1}{4} \left(\int_0^\infty \frac{\cos x}{e^x} dx - \int_0^\infty \frac{\cos 3x}{e^x} dx \right)$$

$$\frac{1}{4} \left(\frac{1}{2} - \frac{1}{10} \right)$$

$$\frac{1}{4} \left(\frac{4}{10} \right)$$

$$\frac{1}{10}$$